Lakshya Tiwari

Mechanical Engineer and PhD Researcher, The University of Texas at Dallas

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Summary

Mechanical engineer and PhD researcher developing a laser-assisted process for joining silicon carbide (SiC) ceramics for high-temperature structural applications. Two years of medical device R&D in an FDA-regulated, ISO 13485 environment, plus research in additive manufacturing and metal–polymer printing, with five publications.

Education

PhD, Mechanical Engineering Jan 2026 – Dec 2030 (expected)

The University of Texas at Dallas, Richardson, TX

Research: laser-assisted joining of silicon carbide ceramics

MS, Aerospace Engineering Aug 2021 – May 2023

Arizona State University, Tempe, AZ · GPA 3.37/4.0

Thesis: Physical modeling and simulation of polymeric structures with metallic material printed by electrically assisted vat photopolymerization (Advisor: Prof. Xiangjia Li)

BTech, Aerospace Engineering Aug 2016 – May 2020

SRM Institute of Science and Technology, Chennai, India · GPA 7.56/10

Research Experience

Research Assistant Jan 2026 – Present

The University of Texas at Dallas, Dallas, TX

- Set up and operate a diode laser joining system integrated with an industrial robot, including I/O integration and motion programming for automated, repeatable laser sequences.
- Develop joining compositions through design of experiments (DOE), screening formulations for wetting, bond quality and thermal stability.
- Fabricate interlayers by tape casting (doctor blade), controlling thickness and uniformity for consistent joints.
- Characterize joints through shear testing, thermal cycling and fractography to link process parameters to joint strength and failure modes.
- Built lab documentation and logbook workflows for traceable, reproducible experiments, applying design-control practices from regulated medical device work.

Research Assistant Oct 2021 – May 2023

Lab of Manufacturing Innovation, Arizona State University, Tempe, AZ

- Fabricated metal–polymer composite structures by electrically assisted 3D printing; evaluated copper deposition (50–100 V) through SEM/EDS, conductivity and mechanical testing.
- Formulated photocurable resins (0–3% PEDOT:PSS), optimized curing parameters over 10 iterative cycles, and simulated nine designs in COMSOL Multiphysics.
- Processed and purified Sb_2Te_3 by controlled heat pressing (100–200 °C, 500–1000 N); optimized sintering and assessed thermal stability with TGA.
- Streamlined experimental procedures with a multidisciplinary team, reducing setup time by 15%.

Professional Experience

Mechanical Engineer Sep 2023 – Nov 2025

BCVS Group, Inc. · Medical device engineering consultant for clients including Baxter and Smith & Nephew

- Designed and released mechanical components and assemblies from concept through design verification, applying GD&T (ASME Y14.5) and DFM/DFA with suppliers to lower production cost and improve assembly fit.
- Authored and executed test protocols, test method validations (TMV) and Gauge R&R studies; validated fixtures and assemblies under design controls.
- Contributed to risk management per ISO 14971 (DFMEA/PFMEA), NPI and design-change activities within a quality system compliant with ISO 13485:2016 and FDA QMSR (21 CFR 820).

- Maintained Design History Files, BOMs and engineering change orders in PLM, keeping documentation audit-ready across cross-functional teams.
- Performed FEA, lifecycle testing and rapid prototyping (3D printing) to shorten design iterations and improve product durability.

Mechanical Engineer
Jun 2023 – Aug 2023

Solinst Canada Ltd. (remote, Tempe, AZ)

- Designed packer components in SolidWorks through iterative modeling, prototyping and testing to meet functional and dimensional requirements.
- Ran CFD simulations in ANSYS to optimize flow performance and structural integrity.

Engine Overhauling and Dressing Intern
May 2018 – Jul 2018

Air India Limited, Mumbai, India

- Assisted aircraft maintenance engineers with overhaul and assembly of PW4056 compressor modules and CFM56 combustion chambers to aviation safety standards.
- Performed non-destructive testing (ultrasonic, eddy current, magnetic particle) on GE90 engines to detect internal defects.

Publications

1. Tang, T., Nipu, S. M. A., **Tiwari, L.**, Yoo, M., and Li, X. (2025). Effects of polymer matrix structures on metallic growth in electrically assisted vat photopolymerization for heterogeneous metal–polymer printing. *Journal of Micro and Nano Science and Engineering*, 13(4), 041009. ASME.
2. Tang, T., Joralmon, D., **Tiwari, L.**, Ravishekar, R., He, Q., Pham, L., Yang, Y., and Li, X. (2024). Additive manufacturing of polymer matrix composites via direct ink writing process. In *Additive Manufacturing of Polymer-Based Composite Materials*, pp. 203–246. Woodhead Publishing (Elsevier).
3. **Tiwari, L.** (2023). *Physical modeling and simulation of polymeric structures with metallic material printed by electrically assisted vat photopolymerization for property enhancements.* MS thesis, Arizona State University.
4. **Tiwari, L.**, Tang, T., Rong, J., Shan, W., Yang, Y., and Li, X. (2022). Thermoelectric material fabrication using mask image projection based stereolithography integrated with hot pressing. *Journal of Material Science and Technology Research*, 9(1), 105–113.
5. Dubey, P., Pramod, M. Y., **Tiwari, L.**, and Kannan, B. T. (2020). Preliminary aeroacoustic measurements of conditioned jet flow from a circular nozzle. *IOP Conference Series: Materials Science and Engineering*, 912(2), 022024.

Selected Projects

- **Neural-network rocket landing (2022):** trained a neural-network controller in PyTorch on discrete-time rocket dynamics, using automatic differentiation to minimize landing loss.
- **Brake disc optimization (2022):** used ANSYS with MOGA and Latin hypercube response surfaces to reduce volume, stress and temperature while raising natural frequency.
- **CFD of thermofluid systems (2022):** simulated laminar and turbulent pipe flow, buoyancy-driven convection (k-epsilon model) and viscous heating in ANSYS.
- **Supersonic aircraft design (2019):** led a multidisciplinary aircraft design project in CATIA V5 to a Mach 1.8 concept; aerodynamic simulations improved overall performance by 3–4%.

Skills

Materials and processes	Silicon carbide, advanced ceramics, laser processing, tape casting, additive manufacturing (vat photopolymerization, DIW), SEM/EDS, TGA, mechanical testing
Design and analysis	SolidWorks, PTC Creo, CATIA V5, Siemens NX, Fusion 360, AutoCAD, GD&T, ANSYS, COMSOL Multiphysics, Abaqus, DOE
Quality and regulatory	ISO 13485, ISO 14971, FDA QMSR, GMP, DFMEA/PFMEA, TMV, Gauge R&R, design controls, PLM, Lean Six Sigma
Programming	Python, PyTorch, MATLAB, LabVIEW, Minitab
Certifications	Lean Six Sigma Green Belt (AIGPE); Machine Learning Specialization (DeepLearning.AI, Stanford); Robotics; Statistical Thermodynamics; Autodesk CAD/CAM/CAE, Generative Design and Multi-Axis CNC Toolpaths